Exercises: Transforming Between Differential and Integral Equations

These exercises accompany the essay *Transforming Between Differential and Integral Equations*. They are arranged in five parts of increasing depth: mechanical conversions, second-order equations and kernels, the Leibniz rule, boundary value problems and Green's functions, and structural results. The conversion itself is a quick skill; the point of each exercise is the verification step—differentiating back, checking a known solution, or proving something on the integral side that is awkward on the differential side. Do not skip those steps. Hints follow the exercises; solution sketches follow the hints.

1 First-Order Conversions and Picard Iteration

Exercise 1.1. Convert the initial value problem

$$y'(t) = 2t\, y(t), \qquad y(0) = 1$$

into a Volterra integral equation. Then verify directly that $y(t) = e^{t^2}$ satisfies your integral equation, by computing the integral (substitute $u = s^2$ after inserting the candidate solution).

Exercise 1.2. Apply Picard iteration to the integral equation from Exercise 1.1, starting from $y_0(t) \equiv 1$. Compute y_1, y_2, y_3 explicitly and identify the pattern. Which function's Taylor partial sums are you generating, and how many new terms does each iteration contribute?

Exercise 1.3. Convert the general linear initial value problem

$$y'(t) + p(t)\, y(t) = q(t), \qquad y(t_0) = y_0,$$

into a Volterra equation of the second kind, identifying the kernel $K(t, s)$ and the forcing term $g(t)$ in the standard form $y(t) = g(t) + \int_{t_0}^{t} K(t, s)\, y(s)\, ds$. Is the kernel of *convolution type* (a function of $t - s$ alone)? Under what condition on p would it be?

Exercise 1.4. Go the other way. Given the integral equation

$$y(t) = 1 + \int_0^t s\, y(s)\, ds,$$

(a) differentiate to obtain an initial value problem, taking care to state where the initial condition comes from; (b) solve the IVP; (c) substitute the solution back into the integral equation and confirm it holds. Why is step (a) legitimate — that is, what regularity of y do you need, and how does the equation itself supply it?

Exercise 1.5. A student converts $y' = f(t, y)$, $y(0) = y_0$ by integrating over the fixed interval $[0, 1]$ instead of $[0, t]$, obtaining $y(1) - y(0) = \int_0^1 f(s, y(s))\, ds$. Exhibit two *different* functions that both satisfy this relation together with $y(0) = y_0$ for the concrete case $f(t, y) = y$, $y_0 = 1$, thereby demonstrating that the fixed-interval relation fails to determine the solution. (You may allow your second function to be any smooth function agreeing with the correct endpoint values and integral.)

2 Second-Order Equations and the Cauchy Kernel

Exercise 2.1. Convert

$$y'' + y = 0, \qquad y(0) = 0, \quad y'(0) = 1,$$

into a Volterra equation. (Move y to the right-hand side first, so the equation reads $y'' = -y$.) Then verify that $y(t) = \sin t$ satisfies the resulting equation

$$y(t) = t - \int_0^t (t - s)\, y(s)\, ds$$

by direct computation of the integral (integrate by parts twice, or split $(t - s) = t - s$ and integrate each piece).

Exercise 2.2. Derive the $n = 3$ case of Cauchy's repeated-integration formula: show by two applications of Fubini's theorem that

$$\int_{t_0}^{t}\int_{t_0}^{u_2}\int_{t_0}^{u_1} h(s)\, ds\, du_1\, du_2 = \int_{t_0}^{t} \frac{(t-s)^2}{2}\, h(s)\, ds.$$

Then state and prove the general-n formula by induction, identifying exactly where the factor $1/(n-1)!$ accumulates.

Exercise 2.3. Convert the third-order problem

$$y''' = f(t), \qquad y(0) = a, \quad y'(0) = b, \quad y''(0) = c,$$

into a single integral formula for $y(t)$ with no unknowns on the right-hand side. Confirm that each initial condition was harvested by exactly one integration, and identify which polynomial term each one produced.

Exercise 2.4. Convert the Airy equation with initial data,

$$y'' = t\, y, \qquad y(0) = 1, \quad y'(0) = 0,$$

into a Volterra equation, and run two steps of Picard iteration from $y_0(t) \equiv 1$. Compare the terms you generate with the known power series of the Airy-type solution $(1 + \frac{t^3}{6} + \frac{t^6}{180} + \cdots)$.

3 The Leibniz Rule: Differentiating Back

Exercise 3.1. Let K and g be smooth, and suppose y is continuous and satisfies

$$y(t) = g(t) + \int_{t_0}^{t} K(t, s)\, y(s)\, ds.$$

(a) State the Leibniz rule for $\frac{d}{dt}\int_{t_0}^{t} K(t,s)\, y(s)\, ds$, identifying the boundary term and the interior term. (b) Differentiate the equation once. For a general kernel, what type of equation results — differential, or integro-differential? (c) Show that if $K(t, s) = k(s)$ depends only on s, one differentiation yields a genuine first-order ODE.

Exercise 3.2. Specialize Exercise 3.1 to the Cauchy kernel $K(t, s) = t - s$:

$$y(t) = g(t) + \int_{t_0}^{t} (t - s)\, y(s)\, ds.$$

Differentiate twice, carefully evaluating the boundary term at each stage (note that $K(t, t) = 0$ for this kernel — what does that buy you?). Recover a second-order ODE together with *two* initial conditions at t_0, read off from the equation and its first derivative. This is the full inverse of the forward transformation.

Exercise 3.3. Show that a Volterra equation with convolution kernel, $y(t) = g(t) + \int_0^t k(t-s)\,y(s)\,ds$, can be solved formally with the Laplace transform: derive $Y(p) = G(p)/(1-\widehat{k}(p))$. Then apply this to Exercise 2.1 (where $g(t) = t$ and $k(\tau) = -\tau$) and confirm that the transform you obtain is that of $\sin t$. This connects the kernel to the transfer-function picture: identify which object plays the role of the impulse response.

4 Boundary Value Problems and Green's Functions

Exercise 4.1. Construct from scratch the Green's function for

$$-y'' = f(t), \qquad y(0) = y(1) = 0,$$

as follows: (a) for fixed s, solve $-y'' = 0$ separately on $[0, s)$ and $(s, 1]$, imposing the boundary condition on each side; (b) glue the pieces by requiring continuity at $t = s$ and a jump of -1 in y' across $t = s$; (c) verify your result agrees with $G(t, s) = s(1-t)$ for $s \le t$ and $t(1-s)$ for $s \ge t$. Explain, in terms of integrating $-y'' = \delta(t-s)$ across the singularity, where the jump condition comes from.

Exercise 4.2. With G as in Exercise 4.1, verify by direct differentiation (splitting the integral at t) that

$$y(t) = \int_0^1 G(t, s)\, f(s)\, ds$$

satisfies $-y'' = f$ and both boundary conditions, for continuous f. Which single step in the computation uses the jump condition?

Exercise 4.3. For the eigenvalue problem $y'' + \lambda y = 0$, $y(0) = y(\pi) = 0$: (a) write the equivalent Fredholm equation $y(t) = \lambda \int_0^\pi G(t, s)\, y(s)\, ds$, constructing G on $[0, \pi]$ by the method of Exercise 4.1; (b) verify by explicit integration that $y(t) = \sin(nt)$ satisfies it with $\lambda = n^2$, i.e. that

$$\int_0^\pi G(t, s)\, \sin(ns)\, ds = \frac{\sin(nt)}{n^2};$$

(c) interpret the result: which operator has eigenvalues n^2, which has $1/n^2$, and why does compactness of the integral operator force its eigenvalues to accumulate at 0?

Exercise 4.4. Show how inhomogeneous boundary data enter the integral formulation: convert

$$y'' = f(t, y), \qquad y(0) = \alpha, \quad y(1) = \beta,$$

into $y(t) = \ell(t) + \int_0^1 G(t, s)\, f(s, y(s))\, ds$, where ℓ is a first-degree polynomial. Determine ℓ, and compare its role with the role of $y_0 + v_0(t - t_0)$ in the initial-value case: in each setting, what equation does the polynomial part solve, and what data does it carry?

5 Structural Results

Exercise 5.1. (Regularity bootstrap.) Let f be continuous, and let $y \in C[0, T]$ satisfy $y(t) = y_0 + \int_0^t f(s, y(s))\, ds$. Prove carefully, quoting the precise part of the Fundamental Theorem of Calculus you use, that $y \in C^1[0, T]$. Then formulate and prove the corresponding statement for the second-order Volterra form with kernel $(t - s)$: continuity of y implies $y \in C^2$.

Exercise 5.2. (Ill-posedness of the first kind.) Consider the Fredholm equation of the first kind $\int_0^1 K(t,s)\,y(s)\,ds = g(t)$ with a smooth kernel. Let $y_n(s) = \sin(n\pi s)$. (a) Show, using the Riemann–Lebesgue lemma (or integration by parts if K is C^1 in s), that the corresponding data $g_n(t) = \int_0^1 K(t,s)\,y_n(s)\,ds$ tends to 0 uniformly as $n \to \infty$. (b) Conclude that arbitrarily large (unit-size) perturbations of the unknown produce arbitrarily small perturbations of the data, and explain in one paragraph why this violates Hadamard's continuous-dependence requirement for the *inverse* problem. (c) Why does no such difficulty arise for the second-kind equation $y = g + \lambda K y$ when $|\lambda|$ is small?

Exercise 5.3. (Why the analysis lives on the integral side.) On $C[0,h]$ with the supremum norm, define $(Ty)(t) = y_0 + \int_0^t f(s, y(s))\,ds$ where f is continuous and satisfies the Lipschitz bound $|f(s,u) - f(s,v)| \le L|u - v|$. (a) Show $\|Ty - Tz\|_\infty \le Lh\|y - z\|_\infty$, so that T is a contraction when $h < 1/L$. (b) Explain in a sentence or two why no analogous estimate can hold for the differentiation operator—what fails? (c) (Optional refinement.) Show that the *second* iterate satisfies $\|T^2 y - T^2 z\|_\infty \le \frac{(Lh)^2}{2}\|y - z\|_\infty$, and that the n-th iterate carries $(Lh)^n/n!$, so that some iterate is a contraction for *every* h. Where have you seen the kernel $(t-s)^{n-1}/(n-1)!$ before in this exercise set?

Exercise 5.4. (Challenge: a genuinely non-differential kernel.) Abel's integral equation is

$$\int_0^t \frac{y(s)}{\sqrt{t-s}}\,ds = g(t).$$

(a) Explain why the strategy of "differentiate until the integral disappears" fails here: what goes wrong with the boundary term when you apply the Leibniz rule? (b) Verify the solution formula

$$y(t) = \frac{1}{\pi}\frac{d}{dt}\int_0^t \frac{g(s)}{\sqrt{t-s}}\,ds$$

in the special case $g(t) = t$, by computing both sides explicitly. (c) The kernel $(t-s)^{-1/2}$ can be viewed as the case $n = \frac{1}{2}$ of the Cauchy kernel family $(t-s)^{n-1}/\Gamma(n)$. What operation does the equation then ask you to invert?

6 Hints

1.1. The integral equation is $y(t) = 1 + \int_0^t 2s\, y(s)\, ds$. With $y(s) = e^{s^2}$ the integrand is an exact derivative.

1.2. Each iteration integrates the previous polynomial term by term; $\int_0^t 2s \cdot \frac{s^{2k}}{k!}\, ds = \frac{t^{2(k+1)}}{(k+1)!}$.

1.3. Move $p(t)y$ to the right side before integrating. The kernel is $-p(s)$; a kernel independent of t is trivially of convolution type only if it is constant.

1.4. FTC Part 1 needs a continuous integrand; continuity of y is given by hypothesis and continuity of $s\, y(s)$ follows. The initial condition is the equation evaluated at $t = 0$.

1.5. The true solution is e^t, with $\int_0^1 e^s ds = e - 1$ and $y(1) = e$. Now take any smooth ϕ with $\phi(0) = 1$, $\phi(1) = e$, $\int_0^1 \phi = e - 1$: three linear conditions, infinitely many smooth functions. A cubic polynomial ansatz works.

2.1. For the verification, compute $\int_0^t (t - s) \sin s\, ds = t - \sin t$.

2.2. The domain of the triple integral is $\{t_0 \le s \le u_1 \le u_2 \le t\}$; integrate out u_1 then u_2, or both at once: the volume of the simplex $\{s \le u_1 \le u_2 \le t\}$ in the (u_1, u_2) variables is $(t - s)^2/2$.

2.3. $y(t) = a + bt + \frac{c}{2}t^2 + \int_0^t \frac{(t-s)^2}{2} f(s)\, ds$.

2.4. $y(t) = 1 + \int_0^t (t - s)\, s\, y(s)\, ds$. Note the kernel is $(t - s)s$, not $(t - s)$: the factor t from the equation rides along inside.

3.1. $\frac{d}{dt} \int_{t_0}^t K(t, s)y(s)ds = K(t, t)y(t) + \int_{t_0}^t \partial_t K(t, s)\, y(s)\, ds$.

3.2. $K(t, t) = 0$ kills the boundary term on the first differentiation; after one derivative the kernel is 1, and the second differentiation produces the boundary term $y(t)$.

3.3. $\widehat{k}(p) = -1/p^2$, $G(p) = 1/p^2$, so $Y(p) = \frac{1/p^2}{1+1/p^2} = \frac{1}{p^2+1}$.

4.1. On each side the solution of $-y'' = 0$ is affine; the boundary conditions force $y = At$ on the left and $y = B(1 - t)$ on the right. Continuity and the jump give two equations for A, B.

4.2. Write $y(t) = (1 - t) \int_0^t sf(s)ds + t \int_t^1 (1 - s)f(s)ds$ and differentiate; the terms from the moving limits cancel on the first derivative and produce $-f(t)$ on the second.

4.3. On $[0, \pi]$, $G(t, s) = s(\pi - t)/\pi$ for $s \le t$. Split the integral at t and integrate by parts, or just integrate directly; everything is elementary.

4.4. ℓ solves $\ell'' = 0$ with $\ell(0) = \alpha$, $\ell(1) = \beta$; the polynomial part always carries the data and is annihilated by the differential operator.

5.1. FTC Part 1 for the first claim. For the second, differentiate the $(t - s)$-kernel equation once via Exercise 3.2 and apply the first claim to the result.

5.2. Integration by parts in s produces a factor $1/(n\pi)$ and bounded terms, uniformly in t if $\partial_s K$ is bounded.

5.3. For (b): consider $y_n(t) = \frac{1}{n} \sin(n^2 t)$, which is uniformly small while y_n' is uniformly large. For (c): when you iterate, the inner integral deposits a factor $(t - s)$, then $(t - s)^2/2$, exactly as in Exercise 2.2.

5.4. (a) $K(t, t) = (t - t)^{-1/2}$ is infinite. (b) With $g(s) = s$, substitute $s = t \sin^2 \theta$; the inner integral evaluates to $\frac{\pi}{2}\, t \cdot \frac{4}{3} \cdot \frac{1}{2} \cdot \ldots$ — carry the Beta function through and differentiate. (c) A half-order integral; the equation asks you to invert semi-integration, and applying it twice gives ordinary integration.

7 Solution Sketches

1.1. Integrating $y'(s) = 2s\,y(s)$ over $[0,t]$: $y(t) = 1 + \int_0^t 2s\,y(s)\,ds$. Verification: $\int_0^t 2s\,e^{s^2}ds = \big[e^{s^2}\big]_0^t = e^{t^2} - 1$, so the right-hand side is $1 + e^{t^2} - 1 = e^{t^2}$. ✓

1.2. $y_1 = 1 + t^2$; $y_2 = 1 + t^2 + \frac{t^4}{2}$; $y_3 = 1 + t^2 + \frac{t^4}{2} + \frac{t^6}{6}$. These are the partial sums of $e^{t^2} = \sum t^{2k}/k!$, one new term per iteration.

1.3. $y(t) = \Big(y_0 + \int_{t_0}^t q(s)\,ds\Big) + \int_{t_0}^t \big(-p(s)\big)\,y(s)\,ds$, so $g(t) = y_0 + \int_{t_0}^t q$ and $K(t,s) = -p(s)$. This is a function of $t - s$ alone only when p is constant (then $K \equiv -p$), the autonomous case.

1.4. (a) The map $t \mapsto \int_0^t s\,y(s)\,ds$ has a continuous integrand (product of continuous functions), so FTC Part 1 applies: $y'(t) = t\,y(t)$, and $t = 0$ in the equation gives $y(0) = 1$. (b) Separable: $y = e^{t^2/2}$. (c) $\int_0^t s\,e^{s^2/2}ds = e^{t^2/2} - 1$; adding 1 closes the loop. The regularity needed for (a) — continuity of y — is exactly what the equation presupposes, and differentiability is then derived, not assumed.

1.5. The relation demands $\phi(0) = 1$, $\phi(1) = e$, $\int_0^1 \phi = e - 1$. Besides e^t, seek $\phi(t) = 1 + \alpha t + \beta t^2$: the endpoint condition gives $\alpha + \beta = e - 1$, and the integral condition gives $1 + \frac{\alpha}{2} + \frac{\beta}{3} = e - 1$, i.e. $3\alpha + 2\beta = 6e - 12$. Solving the pair: $\alpha = 4e - 10$, $\beta = 9 - 3e$; one checks $\alpha + \beta = e - 1$. ✓ This quadratic $\phi \neq e^t$ satisfies the fixed-interval relation but not the ODE; a single scalar equation cannot pin down a function.

2.1. Conversion gives $y(t) = t - \int_0^t (t-s)\,y(s)\,ds$ (the t term carries $y'(0) = 1$; the constant term vanishes since $y(0) = 0$). Verification: $\int_0^t (t-s)\sin s\,ds = t\int_0^t \sin s\,ds - \int_0^t s\sin s\,ds = t(1-\cos t) - (\sin t - t\cos t) = t - \sin t$. Then $t - (t - \sin t) = \sin t$. ✓

2.2. The triple integral is over the simplex $t_0 \le s \le u_1 \le u_2 \le t$. Integrating in the order u_1, then u_2: the u_1-integral gives $(u_2 - s)$; the u_2-integral of $(u_2 - s)$ over $[s,t]$ gives $(t-s)^2/2$. Induction: each additional integration integrates $(u-s)^{k-1}/(k-1)!$ in u from s to the new limit, producing $(t-s)^k/k!$ — the factorial accumulates one factor per layer.

2.3. Three integrations harvest c, then b, then a: $y(t) = a + bt + \frac{c}{2}t^2 + \int_0^t \frac{(t-s)^2}{2} f(s)\,ds$. The k-th initial condition (the value of $y^{(k)}(0)$) produces the term $y^{(k)}(0)\,t^k/k!$ — the polynomial part is precisely the degree-2 Taylor polynomial of the solution at 0.

2.4. $y(t) = 1 + \int_0^t (t-s)\,s\,y(s)\,ds$. First iteration: $\int_0^t (t-s)s\,ds = \frac{t^3}{6}$, so $y_1 = 1 + \frac{t^3}{6}$. Second: add $\int_0^t (t-s)s \cdot \frac{s^3}{6}\,ds = \frac{t^6}{180}$, so $y_2 = 1 + \frac{t^3}{6} + \frac{t^6}{180}$, matching the known series.

3.1. (a) Boundary term $K(t,t)\,y(t)$ plus interior term $\int \partial_t K(t,s)\,y(s)\,ds$. (b) In general the derivative of the equation still contains an integral of y against $\partial_t K$: an *integro-differential* equation. (c) If $K = k(s)$ then $\partial_t K = 0$ and the interior term dies: $y'(t) = g'(t) + k(t)\,y(t)$, a genuine ODE.

3.2. First derivative: boundary term $K(t,t)y(t) = 0$, interior term $\int_{t_0}^t 1 \cdot y(s)\,ds$; so $y' = g' + \int_{t_0}^t y$. Second derivative: now the "kernel" is 1, whose boundary term is $y(t)$: $y'' = g'' + y$. Initial data: $y(t_0) = g(t_0)$ from the equation, $y'(t_0) = g'(t_0)$ from its first derivative (the integrals vanish at $t = t_0$). The vanishing of the Cauchy kernel on the diagonal is what delays the appearance of y outside the integral until the second differentiation — one differentiation per order of the original equation.

3.3. Transforming, $Y = G + \widehat{k}\,Y$, so $Y = G/(1 - \widehat{k})$. For Exercise 2.1: $G(p) = 1/p^2$, $\widehat{k}(p) = -1/p^2$, hence $Y = \frac{1/p^2}{1+1/p^2} = \frac{1}{p^2+1} = \mathcal{L}\{\sin t\}$. ✓ The resolvent kernel of the integral equation plays the role of the impulse response; the map $G \mapsto Y$ is multiplication by the transfer function $1/(1-\widehat{k})$.

4.1. Left piece $y = At$ (satisfies $y(0) = 0$), right piece $y = B(1-t)$ (satisfies $y(1) = 0$). Continuity at s: $As = B(1-s)$. Jump: $y'(s^+) - y'(s^-) = -B - A = -1$. Solving: $A = 1-s$, $B = s$, giving $G(t,s) = t(1-s)$ for $t \le s$ and $s(1-t)$ for $t \ge s$. The jump condition comes from integrating $-y'' = \delta(\cdot - s)$ over $[s-\epsilon, s+\epsilon]$: the left side gives $-(y'(s^+) - y'(s^-))$, the right side gives 1.

4.2. With $y(t) = (1-t)\int_0^t sf\,ds + t\int_t^1 (1-s)f\,ds$: the first derivative is $-\int_0^t sf\,ds + (1-t)\,tf(t) + \int_t^1 (1-s)f\,ds - t(1-t)f(t)$; the two boundary contributions cancel (this cancellation *is* the continuity of G). Differentiating again: $-tf(t) - (1-t)f(t) = -f(t)$, i.e. $-y'' = f$ — this step is the jump condition in action. Boundary values: at $t = 0$ the first integral is empty and the prefactor t kills the second; similarly at $t = 1$.

4.3. (a) $G(t,s) = s(\pi - t)/\pi$ for $s \le t$, $t(\pi - s)/\pi$ for $s \ge t$; the Fredholm form is $y = \lambda\int_0^\pi G\,y$. (b) Split at t: $\int_0^t \frac{s(\pi-t)}{\pi}\sin(ns)\,ds + \int_t^\pi \frac{t(\pi-s)}{\pi}\sin(ns)\,ds$. Each piece is elementary; the $\cos(nt)$ terms cancel between the two pieces and the total is $\sin(nt)/n^2$. ✓ (c) $L = -d^2/dt^2$ has eigenvalues $n^2 \to \infty$ (unbounded operator); $K = L^{-1}$ has $1/n^2 \to 0$. A compact operator maps the unit ball to a precompact set, so it cannot have infinitely many eigenvalues bounded away from zero — accumulation at 0 is forced, and it is the mirror image of the unboundedness of L.

4.4. $\ell(t) = \alpha + (\beta - \alpha)t$. In both settings the polynomial part is the solution of the *homogeneous* equation ($\ell'' = 0$, resp. $y'' = 0$) that carries all the auxiliary data, while the integral term is a particular solution with zero data. Superposition splits "data" from "forcing" cleanly in the integral representation.

5.1. The integrand $s \mapsto f(s, y(s))$ is continuous, so FTC Part 1 says the integral term is C^1 with derivative $f(t, y(t))$; since y equals a constant plus this term, $y \in C^1$. For the second-order form, Exercise 3.2 shows $y' = g' + \int_0^t f(s, y(s))\,ds$ (with g the polynomial part, so g' is smooth); the right side is again C^1 by the same argument, so $y \in C^2$. Each differentiation of the equation trades one layer of kernel for one derivative of y.

5.2. (a) Integrating by parts in s: $g_n(t) = \left[-\frac{\cos(n\pi s)}{n\pi}K(t,s)\right]_0^1 + \frac{1}{n\pi}\int_0^1 \cos(n\pi s)\,\partial_s K(t,s)\,ds$, and both terms are $O(1/n)$ uniformly in t if $K, \partial_s K$ are bounded. (b) The inputs y_n have unit sup-norm but the outputs $g_n \to 0$: inverting the map sends the small perturbation g_n of the data 0 to the unit-size perturbation y_n of the solution 0; continuous dependence fails, so the inverse problem is ill-posed in Hadamard's sense even if a solution exists and is unique. (c) For the second kind, $y = g + \lambda K y$ gives $(I - \lambda K)y = g$; for $|\lambda|\,\|K\| < 1$ the Neumann series makes $(I - \lambda K)^{-1}$ a *bounded* operator, so small changes in g produce small changes in y. The identity term is what saves you: you are never inverting the smoothing operator alone.

5.3. (a) $|Ty(t) - Tz(t)| \le \int_0^t L|y(s) - z(s)|\,ds \le Lt\,\|y - z\|_\infty \le Lh\|y - z\|_\infty$. (b) Differentiation admits no bound $\|y'\|_\infty \le C\|y\|_\infty$: the functions $\frac{1}{n}\sin(n^2 t)$ are uniformly small with uniformly large derivatives. (c) Keeping the factor Lt un-relaxed and iterating: $|T^2 y - T^2 z|(t) \le \int_0^t L \cdot Ls\,\|y - z\|\,ds = \frac{(Lt)^2}{2}\|y - z\|$, and inductively $(Lt)^n/n!$ — the Cauchy kernel of Exercise 2.2 reappearing as the engine of global existence: since $(Lh)^n/n! \to 0$, some iterate is a contraction on any interval.

5.4. (a) The Leibniz boundary term requires $K(t,t)$, but $(t-s)^{-1/2}$ blows up on the diagonal; the rule does not apply, and no finite number of differentiations removes the integral. (b) With $g(s) = s$: $\int_0^t \frac{s}{\sqrt{t-s}}\,ds = \frac{4}{3}t^{3/2}$, so the formula gives $y(t) = \frac{1}{\pi}\cdot\frac{d}{dt}\frac{4}{3}t^{3/2} = \frac{2}{\pi}\sqrt{t}$. Check forward: $\int_0^t \frac{2\sqrt{s}/\pi}{\sqrt{t-s}}\,ds = \frac{2}{\pi}\cdot t\cdot B(\frac{3}{2},\frac{1}{2}) = \frac{2}{\pi}\cdot t\cdot\frac{\pi}{2} = t = g(t)$. ✓ (c) Up to the constant $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, the left side is the half-order integral of y; the equation asks you to invert semi-integration (a fractional-calculus operation), which is why it sits genuinely outside the differential-equation world while remaining a perfectly good Volterra equation.